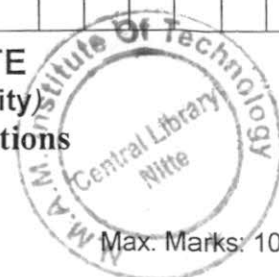


NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
 May - June 2023
CS1003-1 – BASICS OF PYTHON PROGRAMMING



Duration: 3 Hours

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is an invalid assignment?
 A) a,b,c = 10,20,30
 B) a=b=50
 C) a,b,c=12.5,10,"hello"
 D) a b c = 12.5 10 "hello"
- Who developed the Python language?
 A) Zim Den
 B) Guido van Rossum
 C) Niene Stom
 D) Wick van Rossum
- Which among the following belongs to the category of numeric types?
 A) float
 B) str
 C) list
 D) bool
- Which among the following is a logical AND operator in python?
 A) &
 B) &&
 C) and
 D) AND
- Which of the following keywords is used to represent another if statement in python?
 A) elseif
 B) else if
 C) elif
 D) el if
- Which among the following operator can be used with string data types
 A) +
 B) -
 C) *
 D) %
- Which of the following is true for variable in Python
 A) A variable should begin with an underscore
 B) A variable should only have lowercase
 C) A variable is automatically assigned with a specific type based on the assignment
 D) A string data cannot be assigned to the variable
- Suppose list1 is [4, 2, 2, 4, 5, 2, 1, 0], which of the following is correct syntax for slicing operation?
 A) print(list1[0])
 B) print(list1[:2])
 C) print(list1[:-2])
 D) all of the mentioned
- Suppose list1 is [2, 33, 222, 14, 25], What is list1[-1]?
 A) Error
 B) None
 C) 25
 D) 2
- Which of the following is a Python tuple?
 A) [1, 2, 3]
 B) (1, 2, 3)
 C) {1, 2, 3}
 D) {}
- Suppose t = (1, 2, 4, 3), which of the following is invalid?
 A) print(t[3])
 B) t[3] = 45
 C) print(max(t))
 D) print(len(t))
- If t=(1,2,4,3), What will be the output of print([1:3])?
 A) (1, 2)
 B) (1, 2, 4)
 C) (2, 4)
 D) (2, 4, 3)
- If t=(1,2,4,3), What will be the output of print(t[1:-1])?
 A) (1, 2)
 B) (1, 2, 4)
 C) (2, 4)
 D) (2, 4, 3)
- What will following Python code return?
 str1="Stack of books"
 print(len(str1))
 A) 13
 B) 14
 C) 15
 D) 16

15. What will be the output of the following Python code?
- ```
def foo(k):
 k = [1]
 q = [0]
 foo(q)
 print(q)
```
- A) [0] B) [1]  
C) [1, 0] D) [0, 1]
16. How are variable length arguments specified in the function heading?
- A) one star followed by a valid identifier B) one underscore followed by a valid identifier  
C) two stars followed by a valid identifier D) two underscores followed by a valid identifier
17. What is the difference between r+ and w+ modes?
- A) no difference B) in r+ the pointer is initially placed at the beginning of the file and the pointer is at the end for w+  
C) in w+ the pointer is initially placed at the beginning of the file and the pointer is at the end for r+ D) depends on the operating system
18. How do you get the name of a file from a file object (fp)?
- A) fp.name B) fp.file(name)  
C) self.\_\_name\_\_(fp) D) fp.\_\_name\_\_()
19. What happens if no arguments are passed to the seek function?
- A) file position is set to the start of file B) file position is set to the end of file  
C) file position remains unchanged D) error
20. What is the correct syntax of open() function?
- A) file = open(file\_name [, access\_mode][, buffering]) B) file object = open(file\_name [, access\_mode][, buffering])  
C) file object = open(file\_name) D) none of the mentioned

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I  |                                                                                                                                                                           | Marks | BT* | CO* | PO* |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.        | a) Write a Python program to calculate the area of a circle. The program should prompt the user to enter the radius of the circle and then calculate and output the area. | 8     | L2  | 1   | 1   |
|           | b) List and explain the features of the Python programming language.                                                                                                      | 8     | L1  | 1   | 1   |
| 2.        | a) Explain the syntax of the if-else statement and for loop with a programming example for both.                                                                          | 8     | L2  | 1   | 1   |
|           | b) Write Python Program to reverse a number and also find the Sum of digits in the reversed number. Prompt the user for input.                                            | 8     | L3  | 1   | 1   |
| 3.        | a) What is a variable and what are data types? Illustrate the naming conventions for variables.                                                                           | 4     | L1  | 1   | 1   |
|           | b) Develop a program to show the four different uses of the range function with for loop.                                                                                 | 8     | L2  | 1   | 1   |
|           | c) Explain how random numbers are generated in python. Write a suitable code snippet.                                                                                     | 4     | L2  | 1   | 1   |
| Unit – II |                                                                                                                                                                           |       |     |     |     |
| 4.        | a) Write three Python functions to find the Mean, Variance, and Standard Deviation for a list of numbers. Read the numbers for the list from the user.                    | 8     | L3  | 3   | 1   |
|           | b) Develop a Python program to perform a linear search for a given Key number in the list and report Success or Failure. Read the values for the list from the user.      | 8     | L3  | 3   | 1   |
| 5.        | a) Develop a recursive python function to find the factorial of a given number. Read the input from the user.                                                             | 8     | L3  | 3   | 1   |
|           | b) Discuss the relation between tuples and lists, tuples, and dictionaries in detail.                                                                                     | 8     | L2  | 3   | 1   |

6. a) Develop an Object-oriented Python Program to input information of student as given below:  
 a. Name  
 b. Registration Number  
 c. Total and average Marks  
 Write a function to display the values after reading all the values.  
 b) Explain the different types of parameter passing methods. Illustrate with suitable examples.

|   |    |   |   |
|---|----|---|---|
| 8 | L3 | 3 | 1 |
| 8 | L2 | 3 | 1 |

**Unit – III**

7. a) Explain the different methods available for manipulating the CSV files. Show the usage with code snippets.  
 b) Write a Python program to count the total number of vowels and consonants in a String.  
 8. a) Design a python program to read the text input from the user and write the contents to a text file. Read the contents and display them on the screen.  
 b) Explain the use of join() and split() string methods with examples. Describe why strings are immutable with an example.

|   |    |   |   |
|---|----|---|---|
| 8 | L2 | 5 | 1 |
| 8 | L3 | 2 | 1 |
| 8 | L3 | 5 | 1 |
| 8 | L2 | 2 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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